

Three-Family Fermion Mass Ratios from a Closed-Wave Algebraic Model

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Abstract

We propose one closed-cycle mass-ratio law for each three-generation fermion sector:

$$z_{f,k} = 1 + \rho_f \cos\left(\frac{2\pi k}{3} - \frac{2}{9}\right), \quad m_{f,k} = M_f |z_{f,k}|^{2p_f}, \quad k = 0, 1, 2. \quad (1)$$

The three values of k are the three generations. The dimensionless number $z_{f,k}$ is an eigenvalue of a Hermitian 3×3 matrix that cyclically permutes them. The sector scale M_f cancels from mass ratios; ρ_f sets the relative wave strength and p_f stretches the three logarithmic mass separations. Shulga's compact-cycle construction gives the charged-lepton row. We keep the same three-point cycle and phase for the other rows and propose

$$(\rho, p) = (\sqrt{2}, 1), \quad (\sqrt{54/19}, \sqrt{27/19}), \quad (\sqrt{54/35}, \sqrt{5}), \quad (-\sqrt{3}, \sqrt{6})$$

for charged leptons, down quarks, up quarks, and normally ordered neutrinos ($m_1 < m_2 < m_3$). One cubic response, $A \mapsto A - A^3$, converts balanced lepton, charge-weighted quark, and symmetric neutral inputs into the weights $1/2$, $35/54$, $19/54$, and $1/3$; inverse metric normalization gives the four magnitudes $|\rho_f|$. A second postulate chooses p_f by comparing two possible changes of a family mass pattern: changing it relative to a generation-independent background, when such a background exists, or rotating the complete pattern in generation space. The resulting squared lengths are 1, $27/19$, 5, and 6. The negative neutrino sign follows only if its family coupling uses the inner products of the three $SU(3)$ root directions. The two equations form a common action candidate; the source algebras fix the leading cubic response but do not exclude higher invariant terms or force the metric and grading choices in the length selector. At $\mu = M_Z$, the four quark ratios have $\chi^2 \simeq 1.63$ when correlations are ignored. Conditional on an additional two-generation mixing texture and a 90° relative phase, the light-quark ratios give $|V_{us}| = 0.225325$, compared with 0.2251 ± 0.0008 . The framework does not determine absolute mass scales, complete mixing matrices, or gauge- and Higgs-boson masses. A final audit proves which parts of the proposed action follow after stated physical premises and which cannot be selected by the present symmetries alone.

1 Introduction: from Shulga to four sectors

Shulga's charged-lepton equation is

$$m_{\ell,k} = M_\ell \left[1 + \sqrt{2} \cos\left(\frac{2\pi k}{3} - \frac{2}{9}\right) \right]^2, \quad k = 0, 1, 2. \quad (2)$$

Here $k = 2, 1, 0$ correspond to e, μ, τ . The relative wave strength is $\sqrt{2}$, mass is the squared overlap, and a normalized Fourier-mode sum on the three-point cycle gives the phase $2/9$ [1].

Equation (1) keeps those three samples and that phase but permits the wave strength ρ_f and log-mass stretch p_f to depend on the sector. The complete candidate is

sector	ρ	p
e, μ, τ	$\sqrt{2}$	1
d, s, b	$\sqrt{54/19}$	$\sqrt{27/19}$
u, c, t	$\sqrt{54/35}$	$\sqrt{5}$
ν_1, ν_2, ν_3	$-\sqrt{3}$	$\sqrt{6}$

(3)

For the charged-lepton and quark rows, $k = 2, 1, 0$ label the light, middle, and heavy states. For neutrinos, $k = 0, 1, 2$ label ν_1, ν_2, ν_3 . The four unknown scales M_f cancel from ratios.

The ingredients have different status. Shulga derives the charged-lepton formula from a wave on a three-point closed cycle. Division-algebraic models show how three copies of the Standard Model fermion representations can arise from one algebraic construction [2, 3]. Work based on the exceptional Jordan algebra supplies cubic operations and distinguishes up-type from down-type states [4, 5]. Finally, Baez studies an embedding of the Standard Model algebra in the exceptional Lie algebra E_7 . The transformations in E_7 that commute with the Standard Model contain one $SU(3)$ acting on generation labels, and the right-handed neutrino states occupy six off-diagonal directions of that algebra [6]. Here E_7 is used as an organizing algebra; we do not assume that it is an observed gauge symmetry.

Those sources motivate the available three-family structures but do not derive the four rows of Eq. (3). The new proposal is to apply one carrier-response rule and one length-selection rule to those structures. Each rule and each remaining assumption are stated where they enter below.

2 The closed-cycle operator

Let

$$S = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad S^3 = I, \quad (4)$$

so that S sends the first family position to the second, the second to the third, and the third back to the first. It shifts generation labels; it does not describe one particle changing into another generation. Define

$$Z_f = I + \frac{\rho_f}{2} \left(e^{-i\gamma} S + e^{i\gamma} S^\dagger \right), \quad \gamma = \frac{2}{9}. \quad (5)$$

It is useful to name the varying part,

$$T_\gamma = \frac{1}{2} \left(e^{-i\gamma} S + e^{i\gamma} S^\dagger \right), \quad Z_f = I + \rho_f T_\gamma. \quad (6)$$

The discrete Fourier transform diagonalizes every cyclic 3×3 matrix. Applied to S , it gives

$$\text{spec}(Z_f) = \left\{ 1 + \rho_f \cos\left(\frac{2\pi k}{3} - \gamma\right) : k = 0, 1, 2 \right\}.$$

Thus Eq. (1) is the spectrum of $M_f |Z_f|^{2p_f}$, not only a wave analogy. In ordinary matrix language, Z_f is Hermitian and circulant: it is unchanged when all three family labels are shifted together. It

also lies inside the larger Jordan and exceptional-algebra constructions used by the cited sources, but no exceptional multiplication is needed to obtain its spectrum. Its elementary invariants are

$$\text{tr } Z_f = 3, \quad \text{tr}(Z_f - I)^2 = \frac{3}{2}\rho_f^2, \quad \det Z_f = 1 - \frac{3}{4}\rho_f^2 + \frac{1}{4}\rho_f^3 \cos(3\gamma). \quad (7)$$

Only the total loop phase 3γ occurs in the characteristic polynomial.

For a suitable Standard Model embedding in E_7 , Baez finds a unique generation $\mathfrak{sl}_3$, the complexified algebra associated with $SU(3)$ [6]. Here “generation algebra” means that this $SU(3)$ acts on the three family labels while commuting with the Standard Model transformations. Its Weyl symmetry contains a 120° operation that cycles the three weight directions; in Lie-algebra terminology this is an order-three Coxeter element. The matrix S is the simplest representation of that cycle. Baez derives the generation algebra, but identifying this particular discrete operation with the mass cycle is an assumption.

Shulga obtains the offset by evaluating the normalized response

$$\gamma_{C_3} = \sum_{n \neq 0} \frac{|1 - e^{-2\pi i n/3}|^2}{(2\pi n)^2} = \frac{2}{9}. \quad (8)$$

Each integer n labels a Fourier mode around the cycle. The numerator measures how much that mode changes after one of the three equal steps, and the denominator is its one-dimensional Green-function weight. The sum contains no sector charge or exponent. The phase is therefore shared if all four sectors use one cycle and one normalization. That shared-cycle premise is assumed; conditional on it, $2/9$ is not fitted separately.

Finally, Shulga identifies mass with squared overlap. We retain that rule for charged leptons and generalize it to $|Z_f|^{2p_f}$. The exponent multiplies every log-mass difference,

$$\log \frac{m_{f,i}}{m_{f,j}} = 2p_f \log \left| \frac{z_{f,i}}{z_{f,j}} \right|. \quad (9)$$

The action below identifies this stretch with an induced internal length.

3 Carrier response and internal length

3.1 Selecting the wave strength

We first introduce a small auxiliary space that records how one unit of internal weight is divided among the alternatives relevant to a sector. It has two entries for charged leptons and quarks and three entries for neutrinos. This “carrier” is a bookkeeping device, not a new particle state. Let q_f be its initial nonnegative weights, normalized so that their sum is one, and let r_f be the number of entries. Remove the equal-share part by defining

$$A_f = r_f \text{diag}(q_f) - I, \quad \mathcal{F}(A) = A - A^3 + \frac{\text{tr } A^3}{r_f} I. \quad (10)$$

The last term makes $\mathcal{F}(A)$ traceless, so the response redistributes weight without changing the total. The response $A - A^3$ is the lowest odd polynomial that transmits a small imbalance, vanishes when the complementary coherence vanishes, and saturates on a pure two-state carrier. It is also the positive Jordan-triple response of that carrier and the restriction of the cubic gradient of the Freudenthal quartic invariant. These facts fix its relative linear and cubic terms. Unit response near

the balanced state fixes its displayed normalization. Omitting higher invariant corrections remains a leading-order assumption; Appendix H gives the calculation and an explicit counterexample to exact uniqueness.

Let P_f be the proposed final weights, represented by a nonnegative matrix of trace one. We choose it by minimizing the squared mismatch

$$V_{\text{car}}(P_f; A_f) = \frac{1}{2} \|r_f P_f - I - \mathcal{F}(A_f)\|_{\text{HS}}^2. \quad (11)$$

The Hilbert–Schmidt norm is the sum of the absolute squares of the matrix entries. Thus V_{car} is nonnegative and reaches zero exactly when the proposed final weights equal the response. The minimum is

$$P_f = \frac{1}{r_f} [I + \mathcal{F}(A_f)]. \quad (12)$$

The three inputs are

$$q_e = (1/2, 1/2), \quad q_q = (2/3, 1/3), \quad q_\nu = (1/3, 1/3, 1/3). \quad (13)$$

The first assigns equal norm to the two conjugate wave contributions in Shulga’s charged-lepton construction; the entries are not probabilities for an electron and a positron. The second uses the normalized magnitudes $2/3$ and $1/3$ of the up- and down-quark electric charges. This does not identify charge with mass; using charge to initialize the mass carrier is an assumption. The last assigns equal weight to the three neutral family directions described in Sec. 6. Equations (12) and (13) give

$$P_e = \text{diag}(1/2, 1/2), \quad P_q = \text{diag}(35/54, 19/54), \quad P_\nu = I/3. \quad (14)$$

If the quadratic length assigned to one wave coordinate is $w_f |\rho_f|^2$, requiring that coordinate to have unit length gives $w_f |\rho_f|^2 = 1$. We therefore set

$$|\rho_f| = w_f^{-1/2}. \quad (15)$$

This yields every magnitude in Eq. (3). The inverse square root follows exactly if w_f is the metric coefficient of one canonically normalized wave coordinate. Identifying the carrier entry with that metric coefficient is still a physical premise; if it instead measured two endpoint residues, the power of w_f would differ. This distinction is made explicit in Appendix H.

3.2 Interpreting the exponent as a length

The exponent is treated as an internal path length. For a logarithmic amplitude $a(s)$, a fixed cycle eigenvalue z , and an auxiliary path coordinate s , take

$$I[a; z] = \frac{1}{2} \int_0^{L_f} \left(\frac{da}{ds} - \log |z| \right)^2 ds \quad (16)$$

This is another squared-mismatch rule: it is minimized when the logarithmic amplitude changes at the constant rate $\log |z|$. Its endpoint then obeys $a(L_f) - a(0) = L_f \log |z|$. Squaring the endpoint overlap gives

$$\frac{m}{M} = \exp(2[a(L_f) - a(0)]) = |z|^{2L_f}, \quad p_f = L_f. \quad (17)$$

The coordinate s is not time or distance in ordinary space. Equation (16) shows how an internal length can produce the power in the mass law, but it does not derive that internal length from a four-dimensional field theory.

3.3 Choosing among two possible family lengths

It remains to determine L_f . Let $V \simeq \mathbb{C}^3$ be the vector space whose three coordinates label the generations. The four mass patterns do not use this space in the same way:

$$R_e = R_d = \text{End}(V) = \mathbf{1} \oplus \mathbf{8}, \quad R_u = \text{Sym}^2 \bar{V} = \bar{\mathbf{6}}, \quad R_\nu = \mathfrak{sl}(V) = \mathbf{8}. \quad (18)$$

For charged leptons and down quarks, R_f is the space of all 3×3 matrices. It splits into the identity matrix, denoted $\mathbf{1}$, and the eight-dimensional space of traceless matrices, denoted $\mathbf{8}$. The up matrix is symmetric, so it has six independent complex entries and transforms as $\bar{\mathbf{6}}$. The neutrino construction uses the eight generators of $SU(3)$ itself, its adjoint $\mathbf{8}$. The reasons for the up and neutrino assignments are given in Secs. 5 and 6.

All lengths are calculated using the ordinary trace norm inherited from V . To determine whether a family pattern contains a generation-independent background, average it over every family rotation:

$$\Pi_0^{(R)} = \int_{SU(3)} R(g) dg, \quad Q_0 = 1 - \Pi_0 \quad (19)$$

The normalized group measure dg assigns total weight one to $SU(3)$. Anything that changes with orientation cancels in the average. Thus $\Pi_0 Y$ is the invariant, or singlet, part of a pattern Y , while $Q_0 Y = Y - \Pi_0 Y$ is the changing part.

The selector compares two candidate squared lengths. The first is the size of the changing part relative to an invariant background. It is available only if $\Pi_0 Y_f \neq 0$. The second is the total response of the complete sector pattern to one normalized $SU(3)$ family rotation.

To write both candidates in one expression, choose a nonzero trial pattern $Y_f \in R_f$. Let b scale the first candidate, let $X \in \mathfrak{su}(3)$ generate a family rotation, and let $\ell_{R_f}(X)$ be the corresponding generator acting on every index of R_f . The proposed action is

$$\boxed{\mathcal{S}_f(b, X, \lambda) = \frac{1}{2} \left[b^2 \|Q_0 Y_f\|_{R_f}^2 + \|\ell_{R_f}(X)\|_{\text{HS}}^2 \right] - \frac{\lambda}{2} \left[b^2 \|\Pi_0 Y_f\|_{R_f}^2 + \|X\|_V^2 - 1 \right]}. \quad (20)$$

The first line contains the two costs. The term multiplied by λ requires the available reference sizes to add to one; without this constraint both trial coordinates could make the cost zero merely by vanishing. The model assigns opposite parity to the background-change coordinate b and the rotation coordinate X , thereby excluding a cross term between them. This grading, like the action itself, is assumed. The lowest stationary value of the constraint multiplier λ defines $L_f^2 = p_f^2$.

Put $E_f^{\text{rad}} = \|Q_0 Y_f\|^2$ and $N_f^{\text{rad}} = \|\Pi_0 Y_f\|^2$. For the rotation candidate, the ratio of the total squared response in R_f to that in one basic family triplet is independent of which nonzero generator X is used:

$$\frac{\|\ell_{R_f}(X)\|^2}{\|X\|_V^2} = \frac{T(R_f)}{T(\mathbf{3})} =: \kappa_f. \quad (21)$$

The conventional symbol $T(R)$ denotes this trace index. It is not another fitted parameter; it can be obtained by listing how the generator acts on all components of R and summing their squared eigenvalues. Variation of Eq. (20) gives

$$b(E_f^{\text{rad}} - \lambda N_f^{\text{rad}}) = 0, \quad (\kappa_f - \lambda)X = 0. \quad (22)$$

Solving these two stationarity equations gives

$$p_f^2 = \min\left(\frac{E_f^{\text{rad}}}{N_f^{\text{rad}}}, \kappa_f\right) \quad (N_f^{\text{rad}} > 0), \quad p_f^2 = \kappa_f \quad (N_f^{\text{rad}} = 0). \quad (23)$$

If an invariant background exists, the selector compares the two candidate lengths and chooses the smaller. If the group average vanishes, the first candidate has no invariant ruler and the equations set $b = 0$, leaving only the rotation length. Table 1 previews the four applications, which are calculated in the following sections.

Table 1: The two length candidates. A dash means that no invariant background exists, so the radial candidate cannot be normalized.

Sector	Family pattern	Invariant background?	Radial p^2	Rotation p^2
charged lepton	general 3×3 matrix	yes	1	6
down quark	general 3×3 matrix	yes	27/19	6
up quark	symmetric 3×3 matrix	no	—	5
neutrino	$SU(3)$ root pattern	no	—	6

The result is not forced by $SU(3)$ symmetry alone. The action also assumes the grading, the unweighted trace metrics inherited from V , and selection of the lowest stationary channel. Allowing an independent metric weight or a cross term changes the answer; Appendix D gives explicit counterexamples.

4 Charged leptons

Shulga’s balanced real three-component state gives $\rho_e = \sqrt{2}$, while squared overlap gives $p_e = 1$ [1]. The selector action recovers the same exponent. The identity matrix is the generation-independent background, while T_γ is the changing part. Their squared Hilbert–Schmidt lengths are

$$\text{tr } I^2 = 3, \quad \text{tr } T_\gamma^2 = \frac{3}{2}, \quad (24)$$

the wave-to-reference length of $I + \rho_e T_\gamma$ is

$$p_e^2 = \frac{\|\rho_e T_\gamma\|_{\text{HS}}^2}{\|I\|_{\text{HS}}^2} = \frac{\rho_e^2}{2} = 1. \quad (25)$$

The alternative is to rotate the entire 3×3 matrix pattern. The normalized trace response of that full endomorphism space is $\kappa_e = 6$; Eq. (23) therefore selects the smaller value, $\min(1, 6) = 1$. Appendix B shows how all three rotation values 6, 5, 6 are counted. Equation (1) then predicts

$$\frac{m_\mu}{m_e} = 206.7703, \quad \frac{m_\tau}{m_\mu} = 16.8180. \quad (26)$$

The fractional differences from the measured pole-mass ratios are about 10^{-5} . They are larger than the experimental errors, so the relation is close but not exact. It is closely related to Koide’s three-family relation [7].

5 Quarks

5.1 A cubic transfer of normalized charge weights

Write $x_f = 1/\rho_f$, so x_f^2 is the carrier weight assigned to sector f . Two measured mass ratios determine two numerical pairs (ρ, p) , but the inverse problem has two branches. For each quark sector they are

	charge-adjacent	other	
d	(1.68437, 1.19379)	(0.99380, 1.93492)	for (ρ, p) .
u	(1.24282, 2.23404)	(1.60558, 1.76265)	

(27)

The column called “charge-adjacent” is the branch whose carrier weights lie near the normalized charge magnitudes $2/3$ for up and $1/3$ for down. Only this pair also gives nearly complementary fitted weights, $x_u^2 + x_d^2 = 0.999885 \pm 0.002795$. The construction below predicts exact complementary weights close to that branch without using the measured ratios.

Use the normalized magnitudes of the two quark charges as the starting carrier,

$$|q\rangle = \begin{pmatrix} \sqrt{2/3} \\ \sqrt{1/3} \end{pmatrix}, \quad |q\rangle\langle q| = \frac{1}{2}(I + C\sigma_x + D\sigma_z), \quad (28)$$

where

$$D = \frac{1}{3}, \quad C = \frac{2\sqrt{2}}{3}, \quad D^2 + C^2 = 1.$$

The two-component vector is not a physical superposition of up and down quarks. It only packages the two normalized positive weights. The Pauli matrix form separates their diagonal difference D from the off-diagonal coherence C . Charge is not mass, and treating these charge magnitudes as the initial mass carrier is an assumption.

Let $A = D\sigma_z$ be the diagonal up/down asymmetry. Applying the general carrier rule in Eq. (10) reduces here to the cubic polynomial

$$A \mapsto A' = A - A^3. \quad (29)$$

Since $A^2 = D^2 I$,

$$A' = D(1 - D^2)\sigma_z = DC^2\sigma_z = \frac{8}{27}\sigma_z. \quad (30)$$

The new diagonal weights are therefore

$x_u^2 = \frac{1 + 8/27}{2} = \frac{35}{54}, \quad x_d^2 = \frac{1 - 8/27}{2} = \frac{19}{54}.$

(31)

The transferred amount is fixed by the same map:

$$\epsilon = \frac{D - (D - D^3)}{2} = \frac{D^3}{2} = \frac{1}{54}. \quad (32)$$

The numerical identity can also be compared with an earlier exceptional-Jordan result. That construction has squared spread $\Delta = 3/8$ [4]. For that value and the closed-cycle phase, the identity

$$\frac{1}{54} = \frac{3}{8} \left(\frac{2}{9} \right)^2$$

reproduces the earlier expression $\Delta\gamma^2$. This agreement does not derive the cubic response: symmetry still permits higher odd polynomials. Appendix C compares two nearby transfer laws that fit the mass ratios differently.

Inverting Eq. (31) gives

$$\rho_u = \sqrt{\frac{54}{35}}, \quad \rho_d = \sqrt{\frac{54}{19}}. \quad (33)$$

The light down overlap is $z_{d,2} = -0.14398$. It lies beyond a node of the signed wave. The mass remains real because Eq. (1) uses $|z|^{2p_d}$, although a noninteger power is not analytic at the node.

5.2 Different family tensors for the two quark sectors

Let $V \simeq \mathbb{C}^3$ again denote the generation labels. A down-type Yukawa matrix is an ordinary linear map from V to itself, so its family dependence belongs to

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}, \quad (34)$$

the familiar split of a 3×3 matrix into its trace and eight traceless components. This is the natural home of the Hermitian circulant Z_d .

The up-type matrix has an additional symmetry. In a minimal $SU(5)$ description, the Standard Model fields Q , u^c , and e^c of each generation are packaged in a representation conventionally named $\mathbf{10}$. The up coupling contains two identical matter multiplets, $\mathbf{10}_i \mathbf{10}_j \mathbf{5}_H$, where i, j are generation labels. The gauge-index contraction is symmetric under $i \leftrightarrow j$, so the family coefficients obey $Y_u^{ij} = Y_u^{ji}$. A symmetric 3×3 matrix has six independent entries and transforms as

$$Y_u \in \text{Sym}^2 \bar{V} = \bar{\mathbf{6}}. \quad (35)$$

The bar records how the generation indices transform; it does not change the dimension. Appendix A introduces the larger E_7 representations and verifies that they contain both this symmetric family tensor and the required $SU(5)$ Higgs component. That branching calculation is an existence check, not a derivation of the matrix entries.

The same three cycle eigenvalues can nevertheless be placed in either matrix space. Let

$$F_{jk} = \frac{1}{\sqrt{3}} \exp\left(\frac{2\pi i j k}{3}\right), \quad D_z = \text{diag}(z_0, z_1, z_2). \quad (36)$$

Then

$$Z_d = F D_z F^\dagger, \quad Y_u = F D_z F^T = Z_u P, \quad P = F^2. \quad (37)$$

Z_d is Hermitian, while $Y_u^T = Y_u$ is symmetric. Fermion masses are determined by the singular values of a Yukawa matrix—the square roots of the eigenvalues of $Y_u Y_u^\dagger$ —not generally by its signed eigenvalues. Here those singular values are

$$\text{singularvalues}(Y_u) = (|z_0|, |z_1|, |z_2|). \quad (38)$$

Thus the two quark rows can share the closed-cycle spectrum without sharing the same tensor transformation law. This supplies a structural reason for sector-dependent (ρ, p) , but it does not by itself determine their values. Appendix A verifies that the proposed up-sector contraction is nonzero and explains why its overall group-theory coefficient changes only the common up-sector scale, not its mass ratios.

5.3 Trace-metric exponents

The down operator belongs to the same endomorphism space as the charged lepton operator. Equations (24) and (33) therefore give

$$p_d^2 = \frac{\|\rho_d T_\gamma\|_{\text{HS}}^2}{\|I\|_{\text{HS}}^2} = \frac{\rho_d^2}{2} = \frac{27}{19}. \quad (39)$$

It has the same invariant identity background and the same full-matrix rotation value 6 as the charged-lepton operator. The action therefore selects the smaller radial value 27/19. The earlier down-lepton reflection is compatible with this identity but is not needed to obtain it.

The up-family tensor instead consists of symmetric pairs of generation indices, $\bar{\mathbf{6}} = \text{Sym}^2 \mathbf{3}$. To measure how the full paired pattern responds to a family rotation, take

$$X = \text{diag}(1/2, -1/2, 0), \quad \text{tr}_{\mathbf{3}} X^2 = \frac{1}{2}. \quad (40)$$

On a single triplet, X assigns the three values $1/2, -1/2, 0$. On a pair, the two values add. The six symmetric pairs therefore have values $1, -1, 0, 0, 1/2, -1/2$, whose squares sum to $5/2$. Hence

$$p_u^2 = \frac{\text{tr}_{\bar{\mathbf{6}}} X^2}{\text{tr}_{\mathbf{3}} X^2} = \frac{T(\bar{\mathbf{6}})}{T(\mathbf{3})} = 5. \quad (41)$$

No nonzero symmetric two-index tensor remains fixed under every $SU(3)$ family rotation, so $\Pi_0^{(\bar{\mathbf{6}})} = 0$. The rotation channel is therefore the only normalized candidate. The matrix $P = F^2$ in Eq. (37) is a reference only after a family basis has been chosen; averaging it over all $SU(3)$ orientations gives zero, so it cannot act as an invariant background. The result is independent of the chosen nonzero X . It is the ratio of the $SU(3)$ trace indices, with $T(\mathbf{3}) = 1/2$ and $T(\bar{\mathbf{6}}) = 5/2$ [8]. Thus $\sqrt{5}$ comes from counting the squared response of the six symmetric family pairs, not from a fitted ratio or a monomial count vector. The general trace-index count is collected in Appendix B; metric alternatives and their failures appear in Appendix D.

6 Neutrinos

Neutrinos have no measured absolute mass scale. Their family assignment in this model comes from the internal generation $SU(3)$ rather than from electric charge. The algebra $\mathfrak{sl}_3$ has eight generator directions. Two are diagonal; the other six are the off-diagonal matrices E_{ij} with $i \neq j$. These six directions are called root spaces and occur in three oppositely oriented pairs. In Baez's construction, the three right-handed neutrinos and their antiparticles occupy these six spaces [6]. The model assigns equal carrier weight to the three unoriented pairs, $P_\nu = I/3$, so Eq. (15) gives $|\rho_\nu| = \sqrt{3}$.

The six off-diagonal generators do not form a complete $SU(3)$ representation by themselves: family rotations also mix in the two diagonal generators. Together all eight form the adjoint representation $\mathbf{8}$. No nonzero vector in the adjoint remains fixed under every $SU(3)$ rotation, so there is no singlet background and the selector retains only the full-pattern rotation length. For the same X used in Eq. (41), its action by commutator on the six off-diagonal generators has eigenvalues $1, -1, 1/2, -1/2, 1/2, -1/2$; the two diagonal generators have eigenvalue zero. Therefore

$$p_\nu^2 = \frac{\text{tr}_{\mathbf{8}} (\text{ad } X)^2}{\text{tr}_{\mathbf{3}} X^2} = \frac{T(\mathbf{8})}{T(\mathbf{3})} = \frac{3}{1/2} = 6. \quad (42)$$

The trace-index calculation fixes only $|\rho_\nu|$. Its sign requires an additional coupling rule. Let w_1, w_2, w_3 be the three weights of the fundamental $SU(3)$ triplet. The three unoriented root lines can be represented by the differences $\beta_1 = w_2 - w_3$, $\beta_2 = w_3 - w_1$, and $\beta_3 = w_1 - w_2$, their Gram matrix is

$$(\langle \beta_i, \beta_j \rangle) = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}. \quad (43)$$

The diagonal entries are the squared root lengths; the negative off-diagonal entries say that distinct cyclic roots meet at an obtuse angle. If the family-dependent part of the neutrino mass operator is proportional to these root inner products, its cross-family term has the opposite sign from its self-term. Combining that orientation with the carrier magnitude gives $\rho_\nu = -\sqrt{3}$. A generic Majorana mass matrix, which couples a neutrino field to its charge-conjugate field, does not require this root coupling or fix this sign. The negative sign therefore remains conditional. With these choices,

$$\frac{m_2}{m_1} = 47.1804, \quad \frac{m_3}{m_2} = 5.90072, \quad \frac{\Delta m_{32}^2}{\Delta m_{21}^2} = 33.8337. \quad (44)$$

Using the measured solar splitting only to set the scale gives

$$(m_1, m_2, m_3) = (0.000183, 0.008656, 0.05108) \text{ eV}, \quad \sum_i m_i = 0.0599 \text{ eV}.$$

Appendix F compares the adjoint trace-index result with a simple exponent-doubling alternative.

7 Mass-ratio comparison and matching scale

Table 2 uses charged-lepton pole masses [9], quark Yukawa ratios at $\mu = M_Z$ [10], and a neutrino oscillation fit [11]. Quark correlations are not included.

Table 2: Predicted mass ratios. Quark reference values are in the $\overline{\text{MS}}$ scheme at $\mu = M_Z$.

Sector	Ratio	Prediction	Reference value / residual	
charged lepton	m_μ/m_e	206.7703	206.7683	(+0.0010%)
charged lepton	m_τ/m_μ	16.8180	16.8177	(+0.0021%)
down quark	m_s/m_d	19.4167	19.8701 ± 0.3661	(-1.24σ)
down quark	m_b/m_s	53.1458	53.2680 ± 0.7559	(-0.16σ)
up quark	m_c/m_u	502.409	505.682 ± 13.738	(-0.24σ)
up quark	m_t/m_c	272.231	271.629 ± 4.714	$(+0.13\sigma)$
neutrino	$\Delta m_{32}^2/\Delta m_{21}^2$	33.8337	32.83 ± 0.91	$(+1.10\sigma)$

The four quark entries give $\chi^2 \simeq 1.63$. This agreement is localized near the electroweak scale. The light-family ratios run little, but the third-family ratios do not. Using the 2024-PDG-input evolution of Ref. [10] gives the scale audit in Table 3.

The lepton comparison is also not a common-scale Yukawa comparison. At M_Z , Ref. [10] gives approximately $y_\mu/y_e = 210.664$ and $y_\tau/y_\mu = 16.986$, rather than the pole-mass ratios in Eq. (26). Shulga explicitly interprets the charged-lepton equation at the pole-mass level [1]. A parent theory must therefore derive the running and threshold map between its internal spectrum and the observables used here.

Table 3: Running quark Yukawa ratios and the fixed model prediction. The χ^2 values ignore correlations.

μ	y_s/y_d	y_b/y_s	y_c/y_u	y_t/y_c	χ^2
M_Z	19.870	53.268	505.682	271.629	1.63
10^3 GeV	19.852	52.276	505.691	277.042	4.17
10^5 GeV	19.821	50.901	506.903	284.397	13.65
10^{16} GeV	19.846	46.977	505.226	307.172	75.75

8 A conditional Cabibbo prediction

Mass eigenvalues do not determine mixing angles. Teli and Singh introduce a Fritzsch-type two-state bridge from adjacent masses to an angle [12]. For $m_i < m_j$, the real symmetric texture

$$\begin{pmatrix} 0 & \sqrt{m_i m_j} \\ \sqrt{m_i m_j} & m_j - m_i \end{pmatrix} \quad (45)$$

has eigenvalues $-m_i, m_j$ and gives

$$\tan \theta_{ij} = \sqrt{\frac{m_i}{m_j}}. \quad (46)$$

This bridge is an additional texture assumption, not a consequence of the closed-wave spectrum.

Apply it only to the first two quark generations. The model predicts

$$t_d = \tan \theta_{12}^d = \frac{1}{\sqrt{19.4166656}} = 0.2269406, \quad t_u = \tan \theta_{12}^u = \frac{1}{\sqrt{502.408944}} = 0.0446140. \quad (47)$$

If the two edge amplitudes have the balanced relative phase $\phi_{12} = \pi/2$, they are in quadrature: one is real and the other purely imaginary in this convention. Their cross term then vanishes, giving

$$|V_{us}|^2 = \frac{t_d^2 + t_u^2}{(1 + t_d^2)(1 + t_u^2)}. \quad (48)$$

The result is

$$\boxed{|V_{us}| = 0.2253252}, \quad (49)$$

compared with 0.2251 ± 0.0008 [10]. The central value would require $\phi_{12} = 90.30^\circ$ in the convention of Ref. [12]. That work derives a local phase relation between paired up- and down-type states and identifies a 90° separation as its balanced orientation. It does not derive how that local phase becomes the phase between the two CKM rotations; that bridge remains assumed here.

No CKM input was used to obtain the two mass ratios. Conditional on Eqs. (45) and (48), the Cabibbo comparison is therefore an independent γ test of the transfer rule. Appendix C shows that it reverses the mass-only preference for the $\tan^2 \gamma$ alternative and favors the cubic $1/54$ transfer. The success does not extend unchanged to V_{cb} ; Appendix E gives the obstruction.

9 Scope and limitations

The model makes seven ratio predictions and one conditional Cabibbo prediction. Better light-quark determinations directly test the cubic transfer. Normal neutrino ordering and $\Delta m_{32}^2/\Delta m_{21}^2 = 33.8337$ test the neutrino row. The scale audit tests whether the quark relations are electroweak matching conditions rather than high-scale identities. The selector also predicts a channel switch for any

future sector whose family pattern is a general 3×3 matrix with an identity background: if $\rho^2/2 > 6$, rotating the complete pattern costs less than changing it relative to that background, and the selected length is $p^2 = 6$.

The four M_f are undetermined, so the model does not explain absolute rest energies. Equations (11) and (16) are a proposed internal action, not a uniquely derived four-dimensional field theory. Appendix H gives an explicit internal-profile completion in which the singular values of the resulting left–right bilinear are exactly $M_f |z_{f,k}|^{2p_f}$. This establishes that the overlap law can be a physical fermion mass law, but does not make that completion unique. Equation (20) removes the manual two-way case distinction, but it assumes that the background-change and family-rotation coordinates have opposite parity, that every tensor length is induced from one unweighted triplet metric, and that the lowest stationary channel is selected. $SU(3)$ symmetry would also permit independent metric weights, which would change the predicted exponents. The E_7 branching shows that the required field types exist but does not select their background orientations or the unique interaction that connects them to the weak Higgs doublet. The neutrino sign requires the root-overlap coupling. The charged-lepton and quark comparisons also use different mass definitions. Appendix H gives continuous, symmetry-preserving counterexamples when the metric or cross-term assumptions are relaxed. Consequently those two choices cannot be derived from the present $SU(3)$ representation content alone. These are independent open problems.

The model also fixes spectra but not relative family frames. If every sector uses exactly the same shift S , all Z_f commute and are diagonalized by the same Fourier matrix. The resulting CKM and PMNS matrices are trivial up to phases and permutations. Nonzero mixing requires misaligned sector frames or noncirculant terms. Gauge bosons and the Higgs do not have three family samples, so Eq. (1) gives no prediction for their masses. Figure 1 summarizes the twelve fermion states. Its angle is a family coordinate, not ordinary space or time; a particle does not rotate into another generation as it propagates.

10 Conclusion

The model represents the three generations by one cyclic 3×3 operator. Shulga’s construction supplies the charged-lepton row and the cycle phase $2/9$; an order-three symmetry of the generation $SU(3)$ inside E_7 is a possible algebraic source of the shift. One proposed cubic carrier response gives the exact weights $1/2$, $35/54$, $19/54$, and $1/3$ and hence the four magnitudes $|\rho_f|$. One proposed length selector then compares change relative to an invariant identity background with rotation of the complete family pattern. It gives $p_e^2 = 1$ and $p_d^2 = 27/19$ from the first route, and $p_u^2 = 5$ and $p_\nu^2 = 6$ from the second. Negative inner products among the three neutrino root directions provide a candidate reason for $\rho_\nu < 0$. No measured mass ratio is used to choose these eight numbers.

The construction remains conditional. The normalized two-state carrier and the leading Freudenthal invariant fix the direction and relative coefficient of the cubic response, while unit small-bias response fixes its overall normalization. They do not exclude higher invariant corrections. The inverse-square rule follows from canonical normalization of one internal coordinate, and the lower stationary length follows if the observed state is the ground mode of a positive internal operator. By contrast, the source algebras do not force the common induced metric, the parity that removes cross terms, or the assumed use of root inner products in the neutrino mass matrix. The matching relations are not scale independent, and the two-state mixing bridge fails in the $(2, 3)$ sector. Deriving the proposed internal action from a common parent field theory, together with relative family frames and the pole-to-running map, is necessary before the numerical pattern becomes a complete theory of fermion mass and mixing. Appendix H states the minimal extra premises and proves the present

non-uniqueness results.

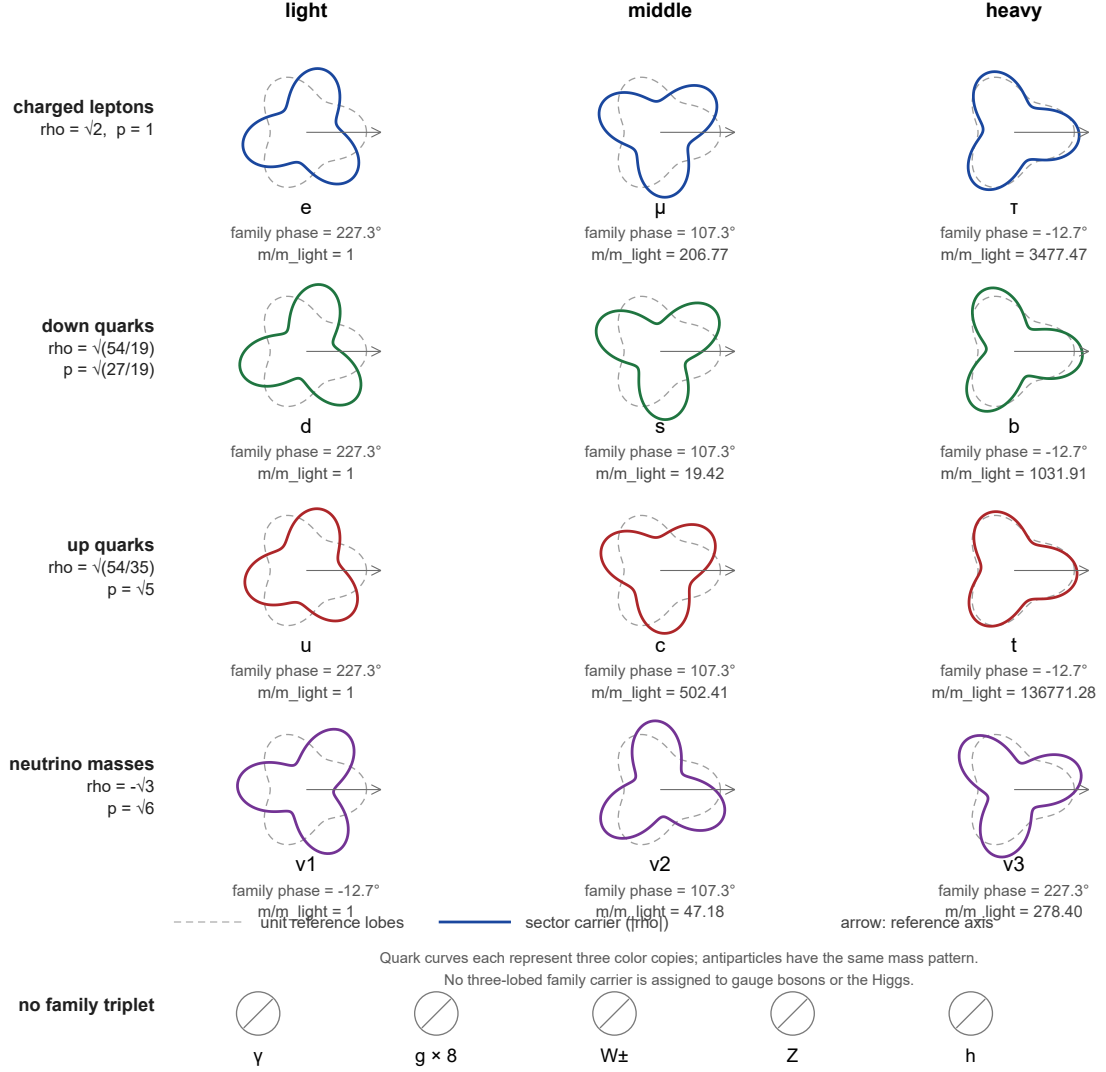


Figure 1: Closed-wave picture of the twelve fermion mass states. Dashed lobes show a unit reference wave. Solid lobes show the sector wave; their radial change is proportional to $|\rho_f|$ and is the same for all three states in a row. Only the family angle changes. Radial distance is a plotting coordinate, not physical size or mass. The neutrino labels are mass states. Each quark curve represents three colors. The light down state has negative signed overlap; its mass uses the absolute value in Eq. (1).

A Explicit up-channel contraction

This appendix checks one limited claim used in Sec. 5.2: the proposed E_7 field content contains a nonzero interaction with a symmetric three-generation tensor and an $SU(5)$ Higgs **5**. It does not calculate a Yukawa matrix or any mass ratio. Bold integers in this appendix name representations by their dimensions; for example, **56** means a 56-dimensional representation, not 56 particles or a numerical coefficient.

A.1 Why the 1463-dimensional representation appears

The starting E_7 matter representation is **56**. The symmetric product of two identical **56** fields has $56 \times 57/2 = 1596$ components and splits into two irreducible E_7 representations—subspaces that cannot be split further while preserving the full E_7 action [8]:

$$\text{Sym}^2(\mathbf{56}) = \mathbf{133} \oplus \mathbf{1463}, \quad 1596 = 133 + 1463. \quad (50)$$

Thus “projection onto the **1463**” means decomposing a symmetric pair according to Eq. (50) and retaining its 1463-dimensional component.

To see why that component is relevant, reduce E_7 to the subgroup $SU(3)_{\text{family}} \times SU(6)$. The **56** contains a component $(\bar{\mathbf{3}}, \mathbf{6})$. The two entries in parentheses say how the same field transforms under the two subgroup factors. Pairing two such components symmetrically gives

$$\text{Sym}^2(\bar{\mathbf{3}}, \mathbf{6}) = (\bar{\mathbf{6}}, \mathbf{21}) \oplus (\mathbf{3}, \mathbf{15}). \quad (51)$$

This follows from separating symmetry in the family indices from symmetry in the $SU(6)$ indices: symmetric–symmetric gives $(\bar{\mathbf{6}}, \mathbf{21})$, while antisymmetric–antisymmetric gives $(\mathbf{3}, \mathbf{15})$. The latter component is already contained in the **133** branch of Eq. (50); the former is therefore in the **1463** branch [6, 8].

The family factor $\bar{\mathbf{6}}$ is exactly the symmetric 3×3 family tensor required for the up Yukawa. The accompanying $SU(6)$ representation $\mathbf{21} = \text{Sym}^2(\mathbf{6})$ also contains the desired $SU(5)$ Higgs representation because

$$\mathbf{6} \downarrow SU(5) = \mathbf{5} \oplus \mathbf{1}, \quad \mathbf{21} \downarrow SU(5) = \mathbf{15} \oplus \mathbf{5} \oplus \mathbf{1}. \quad (52)$$

The downward arrow means that an $SU(6)$ representation is being viewed after restriction to its $SU(5)$ subgroup. Equations (50)–(52) are the entire reason the **1463** enters the paper.

A.2 A nonzero subgroup contraction

It remains to check that the **5** is not merely present in a branching table but can participate in a nonzero interaction. To avoid confusion with the three-dimensional family space V , call the defining vector space of $SU(6)$ $W = \mathbb{C}^6$ and write

$$S \in \text{Sym}^2 W = \mathbf{21}, \quad \Sigma \in \mathfrak{sl}(W) = \mathbf{35}. \quad (53)$$

Here S is a symmetric 6×6 tensor and Σ is a traceless 6×6 generator. The map

$$\boxed{C_\Sigma(S) = \Sigma S - S \Sigma^T} \quad (54)$$

is antisymmetric because transposing it changes its sign, and hence it takes values in $\Lambda^2 W = \mathbf{15}$. It is also equivariant: transforming the inputs and then applying the map gives the same result as applying the map first and transforming its output. Explicitly, under $S \mapsto g S g^T$ and $\Sigma \mapsto g \Sigma g^{-1}$,

$$C_{g \Sigma g^{-1}}(g S g^T) = g C_\Sigma(S) g^T.$$

For two antisymmetric matter tensors $A_i, B_j \in \Lambda^2 W$, a scalar under $SU(6)$ is

$$\mathcal{I}_u = \epsilon_{abcdef} A_i^{ab} B_j^{cd} [C_\Sigma(\mathcal{S}^{ij})]^{ef}, \quad \mathcal{S}^{ij} = \mathcal{S}^{ji}. \quad (55)$$

Here ϵ_{abcdef} is the six-index antisymmetric tensor and i, j are generation labels. Exchanging two 2-forms produces the sign $(-1)^{2 \cdot 2} = +1$, so the contraction is symmetric under interchange of the two up-type matter multiplets, as required.

At the full E_7 level, two background fields Φ are first paired in the **1463** just identified, an adjoint background Σ lies in the **133**, and their product is projected onto a **1539**. The last name again denotes a representation dimension. It is chosen because a pair of fermion fields Ψ has a matching **1539** component, allowing an E_7 scalar contraction:

$$(\Phi\Phi)_{\mathbf{1463}} \otimes \Sigma_{\mathbf{133}} \longrightarrow \mathbf{1539} \longleftrightarrow (\Psi\Psi)_{\mathbf{1539}}. \quad (56)$$

The **1539** occurs once in $\mathbf{1463} \otimes \mathbf{133}$, so there is only one such E_7 contraction apart from an overall normalization. When reduced to $SU(3)_{\text{family}} \times SU(6)$, it contains $(\bar{\mathbf{6}}, \mathbf{15})$ and its conjugate [8]. These are the transformation properties produced by Eq. (54). The subgroup formula is therefore a concrete version of the required part of the E_7 contraction.

To isolate the $SU(5)$ Higgs **5**, choose the traceless diagonal generator

$$Z = \text{diag}(1, 1, 1, 1, 1, -5). \quad (57)$$

For diagonal Z ,

$$C_Z(S)^{ab} = (z_a - z_b)S^{ab}. \quad (58)$$

This generator treats the first five coordinates equally and distinguishes the sixth, so it leaves an $SU(5)$ subgroup unbroken. The components S^{ab} with $a, b \leq 5$, the mixed components S^{a6} , and S^{66} form the **15**, **5**, and **1** in Eq. (52). Consequently,

$$C_Z(S^{ab}) = 0, \quad C_Z(S^{a6}) = 6S^{a6}, \quad C_Z(S^{66}) = 0. \quad (59)$$

The map kills the **15** and **1** but multiplies the **5** by six. The contraction is therefore nonzero and isolates precisely the representation containing the desired weak-doublet component.

With the Hilbert–Schmidt normalization $\hat{Z} = Z/\sqrt{\text{tr } Z^2} = Z/\sqrt{30}$, normalized mixed symmetric and antisymmetric tensors obey

$$C_{\hat{Z}}(S_{a6}) = \frac{6}{\sqrt{30}} A_{a6} = \sqrt{\frac{6}{5}} A_{a6}. \quad (60)$$

This number depends on the generator normalization and does not equal ρ_u , p_u , or p_d . More decisively, it multiplies all generation components of $\mathcal{S}^{ij}$ equally. It changes the overall up-sector mass scale, which cancels from mass ratios. The calculation therefore confirms the up-sector tensor route but supplies no unexplained mass-law parameter. The carrier response and invariant-projector selector in Sec. 3 supply the ratio parameters; the pair-background orientation and the precise operator coupling this channel to the weak Higgs doublet remain open.

B Counting the family-rotation lengths

This appendix derives the rotation values 6, 5, and 6 without assuming prior familiarity with representation indices. Choose the normalized family generator

$$X = \text{diag}(1/2, -1/2, 0), \quad \text{tr}_3 X^2 = 1/2.$$

The trace index ratio $T(R)/T(\mathbf{3})$ is obtained by listing the eigenvalue with which X acts on every component of R , summing their squares, and dividing by $1/2$.

For a general 3×3 matrix, use the nine basis matrices E_{ij} . The family action is the commutator

$$X : E_{ij} \mapsto [X, E_{ij}] = (x_i - x_j)E_{ij}.$$

The three diagonal matrices have eigenvalue zero. The six off-diagonal matrices have eigenvalues

$$1, -1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2},$$

whose squares sum to 3. Therefore

$$\frac{T(\text{End}(\mathbf{3}))}{T(\mathbf{3})} = \frac{3}{1/2} = 6. \quad (61)$$

The identity matrix is one of the zero modes and is fixed by every family rotation. It supplies the invariant background used by the charged-lepton and down-quark radial candidates.

For a symmetric pair of family indices, the generator value is the sum of the two individual values. The six independent pairs have values

$$1, -1, 0, 0, \frac{1}{2}, -\frac{1}{2},$$

whose squares sum to $5/2$. Hence

$$\frac{T(\overline{\mathbf{6}})}{T(\mathbf{3})} = \frac{5/2}{1/2} = 5. \quad (62)$$

No nonzero symmetric pair is invariant under every $SU(3)$ family rotation, so this representation has no singlet background.

Finally, the adjoint $\mathbf{8}$ is the traceless part of the general 3×3 matrix space. Removing the identity removes only a zero mode, so the sum of squared nonzero responses remains 3:

$$\frac{T(\mathbf{8})}{T(\mathbf{3})} = \frac{3}{1/2} = 6. \quad (63)$$

The only matrix commuting with all $SU(3)$ transformations is proportional to the identity, which is not traceless. The adjoint therefore has no invariant singlet background. These elementary counts are the content hidden by the compact notation $T(R)/T(\mathbf{3})$ in the main text.

C Transfer-law ablation with the Cabibbo test

Table 4 keeps the exponent rules fixed and compares the three transfers considered earlier, using the exceptional-Jordan spread $\Delta = 3/8$. A “pull” is the difference between prediction and reference value measured in standard deviations. The last column adds the squared Cabibbo pull from Eq. (49) to the four-mass χ^2 , treating the quoted uncertainties as independent.

Table 4: Mass-ratio and conditional Cabibbo predictions for three transfer laws. The observed Cabibbo input is 0.2251 ± 0.0008 .

Transfer	m_s/m_d	m_b/m_s	m_c/m_u	m_t/m_c	$ V_{us} _{\pi/2}$	joint χ^2
cubic $1/54 = \Delta\gamma^2$	19.4167	53.1458	502.409	272.231	0.225325	1.71
$\Delta \sin^2 \gamma$	19.2639	53.3257	499.909	271.909	0.226160	4.68
$\Delta \tan^2 \gamma$	19.7375	52.7763	507.644	272.899	0.223603	4.15

The $\tan^2 \gamma$ choice gives the smallest mass-only value, $\chi^2 = 0.65$, but its Cabibbo prediction is 1.87σ low. The cubic transfer gives a mass-only value of 1.63 and a Cabibbo pull of 0.28σ . Thus the independent observable reverses the mass-only ranking. This conclusion remains conditional on the two-state bridge and the quadrature phase.

D Exponent-metric ablations

In the cited Jordan-algebra construction, a “Jordan frame” is a set of three mutually orthogonal elementary directions e_1, e_2, e_3 . With the standard trace metric, a combination with integer multiplicities has squared length

$$\|n_1 e_1 + n_2 e_2 + n_3 e_3\|^2 = n_1^2 + n_2^2 + n_3^2.$$

This makes the ordinary Euclidean length of a count vector a natural first candidate, but it does not yield the sector exponents. In the cubic $\text{Sym}^3(\mathbf{3})$ ladders of Refs. [4, 5], a symbol such as $a^2 b$ means that one elementary direction occurs twice and another occurs once. The light down and light up states are both $a^2 b$, with count vector $(2, 1, 0)$. The reflected charged-lepton state is $a^2 c$, with vector $(2, 0, 1)$. Permuting the three elementary directions is a Weyl symmetry, so any invariant trace norm assigns all three vectors the same length $\sqrt{5}$.

It therefore cannot produce $p_e = 1$, $p_d = \sqrt{27/19}$, and $p_u = \sqrt{5}$. This rules out a common monomial-count interpretation of the exponents.

A pure representation-index rule also fails. The charged-lepton and down operators both lie in $\mathbf{1} \oplus \mathbf{8}$, so it would assign them the same length. Conversely, applying the radial relation $p^2 = \rho^2/2$ to all four sectors would give

$$p_u^2 = \frac{27}{35}, \quad p_\nu^2 = \frac{3}{2}, \quad (64)$$

instead of 5 and 6.

The selector action avoids both failures by retaining both trial channels and minimizing only after group projection. On $\mathbf{1} \oplus \mathbf{8}$, the identity has nonzero singlet projection and the radial values 1 and $27/19$ lie below the orbit value 6. On $\bar{\mathbf{6}}$ and $\mathbf{8}$, the singlet projector vanishes and only the orbit values 5 and 6 remain. In particular, after choosing a family frame the symmetric up tensor has a reference bilinear $P = F^2$, but $\Pi_0 P = 0$. The action gives it positive radial cost and zero normalization, so stationarity eliminates that trial channel rather than excluding it by a case statement.

This result is still not forced by representation symmetry alone. If the background-change coordinate and family-rotation coordinate are allowed to have the same parity, the action may contain an invariant cross term between their traceless $\mathbf{8}$ components. Even with the cross term forbidden, one may multiply the trace metric in each representation by a different positive constant without breaking $SU(3)$. Either change alters the selected length. These are counterexamples to a claim of uniqueness based only on $SU(3)$ invariance.

E Mixing obstructions

Every operator

$$Z_f = I + a_f S + a_f^* S^\dagger$$

is a polynomial in the same shift. Hence $[Z_f, Z_g] = 0$ for all sectors and one Fourier matrix diagonalizes them all. Varying ρ_f, p_f , or the coefficient phase cannot by itself generate mixing.

The Cabibbo bridge also cannot be applied unchanged to the (2,3) blocks. The model gives

$$\frac{m_b}{m_s} = 53.1458, \quad \frac{m_t}{m_c} = 272.231.$$

Equation (46) then fixes two rotations. Allowing an arbitrary relative phase, their contribution satisfies

$$0.0757 \leq |V_{cb}| \leq 0.1956, \quad (65)$$

whereas the observed value is about 0.0419 [10]. No phase can repair the discrepancy. A different bridge or a sector-dependent suppression is required, consistent with the phenomenological (2,3) normalization introduced in Ref. [12].

F Neutrino exponent comparison

In a minimal seesaw, a light neutrino mass is approximately a Dirac mass squared divided by a much larger right-handed-neutrino mass. If that heavy mass were identical for all three generations, the light mass ratios would square the Dirac mass ratios. In the present parametrization this would double the exponent, giving $p = 2\sqrt{3}$, rather than the adjoint trace-index value $p = \sqrt{6}$. Keeping $\rho_\nu = -\sqrt{3}$ and $\gamma = 2/9$ fixed gives Table 5.

Table 5: Neutrino predictions from the adjoint trace-index rule and from simple exponent doubling. The latter is not a general seesaw prediction.

Quantity	$p = \sqrt{6}$	$p = 2\sqrt{3}$
m_2/m_1	47.1804	232.839
m_3/m_2	5.90072	12.3091
$\Delta m_{32}^2/\Delta m_{21}^2$	33.8337	150.515

The observed splitting ratio favors the trace-index row over this restricted alternative. Nondegenerate heavy-neutrino masses can produce many other patterns, so the table does not exclude seesaw models generally.

G Plain-language guide to the model

This appendix translates the mathematical model into a physical picture. It is intended as a guide to the equations, not as an additional derivation. Where the picture sounds mechanical, words such as “wave,” “turn,” and “background” refer to the internal family pattern defined in the paper. They do not describe a material object spinning in ordinary space.

G.1 Mass, charge, and spin in this picture

Within the model, mass is energy that remains locked inside a stable particle pattern. A particle with more locked energy takes more energy to create and more energy to accelerate. The same energy also contributes to gravity. Mass can therefore help hold a gravitational orbit together. It does not hold an atomic orbit together by itself; electric attraction supplies that force.

Charge is pictured as a direction and amount of internal twisting. In a Kaluza–Klein picture, motion around a closed hidden direction appears in ordinary space as electric charge. That hidden motion also carries energy, so a charged twist can influence the starting rest energy. Charge is not

equal to mass: opposite twists can have opposite charge but equal energy, and the final energy depends on how the twist combines with the rest of the particle pattern.

Spin describes how a particle responds when ordinary space is rotated. The family turn X in Eq. (20) is not ordinary spin. It turns the three-generation pattern, not the particle in physical space. The present model does not explain why the fermions have spin one half. In a higher-dimensional geometry, ordinary spin and charge could be two different transformation rules of one higher-dimensional field: spin would describe its response to rotations of visible directions, while charge would describe its response to shifts or rotations of compact directions. They would share a geometric origin without being the same rotation. This possibility motivates the twisting language but is not used in the mass-ratio derivation.

The model uses the following chain of ideas. Charge or a neutral symmetry sets a starting balance. The carrier response turns that balance into a wave strength ρ_f . Three allowed family positions then produce cancellation, partial cancellation, and reinforcement. The length selector determines how strongly those three raw differences are enlarged. The mass law converts the resulting overlaps into mass ratios. A separate scale M_f remains unknown for each sector.

G.2 Dictionary of the equations

Table 6: Technical terms and their plain-language meanings. The descriptions state how the proposed model uses each term; they are not definitions accepted independently of the model.

Term	Plain-language meaning
f	The type of fermion: charged lepton, down quark, up quark, or neutrino.
$k = 0, 1, 2$	The three allowed family positions. They label three different stable particles; a particle is not moving from one generation to another.
$\gamma = 2/9$	The common offset of the three-lobed closed wave. It moves all three sampling positions together without changing their equal spacing.
$z_{f,k}$	The signed raw overlap between the steady part and the family wave at one of the three positions. A small absolute value means strong cancellation; a large one means reinforcement.
ρ_f	The strength of the changing family wave compared with the steady part. Its magnitude controls how deep the cancellation and reinforcement can be. Its sign fixes the relative orientation.
p_f	The amount by which the raw three-family differences are enlarged when they are converted into mass differences.
M_f	The unknown overall energy scale of one sector. It shifts all three masses together and cancels from ratios.
$m_{f,k} = M_f z_{f,k} ^{2p_f}$	The complete mass-ratio law. Take the size of the raw overlap, enlarge its differences by p_f , square the result, and then set the sector's overall scale with M_f .
q_f in Eq. (13)	The starting shares assigned to the available carrier directions. Charged leptons use a balanced pair, quarks use the two-to-one magnitudes of their charges, and neutrinos use three equal neutral directions.
r_f	The number of starting carrier directions: two for the charged-lepton and quark balances, and three for the neutrino balance.
$A_f = r_f \text{diag}(q_f) - I$	The starting imbalance after the equal share has been removed. It is zero when all directions begin equally.

Table 6 continued

Term	Plain-language meaning
$\mathcal{F}(A) = A - A^3 + (\text{tr } A^3 / r_f)I$	The model's rule for how the starting imbalance settles. The linear part keeps a small imbalance; the cubic part weakens a large one and produces the quark transfer.
V_{car} , Eq. (11)	The complete carrier-response score. It measures the squared mismatch between the proposed final balance and the balance demanded by the response rule. Its lowest value occurs when the two match exactly.
$r_f P_f - I - \mathcal{F}(A_f)$	The mismatch inside the carrier-response score. It vanishes at the selected carrier balance.
$\frac{1}{2} \ r_f P_f - I - \mathcal{F}(A_f)\ ^2$	A spring-like energy penalty: doubling the mismatch gives four times the penalty.
P_f	The final distribution of one unit of internal capacity among the carrier directions. Its relevant entries are 1/2, 35/54, 19/54, and 1/3.
w_f	The entry of P_f assigned to the sector's wave coordinate.
$ \rho_f = w_f^{-1/2}$	Unit normalization of the wave coordinate. A smaller assigned weight permits a larger wave coordinate, and a larger weight permits a smaller one.
$a(s)$	A running record of the logarithm of the wave size along an internal path. It is a bookkeeping field, not a path through ordinary space.
$I[a; z]$, Eq. (16)	The complete log-amplitude score. It is lowest when the logarithmic wave change follows the value required by the raw overlap z .
$da/ds - \log z $	The local mismatch inside the log-amplitude score.
$L_f = p_f$	The selected internal path length. A longer path repeats or amplifies the logarithmic difference more strongly.
$V = \mathbf{3}$	The basic three-place family space from which all sector patterns are built.
R_f	The way the family pieces are joined in sector f . Different joining rules are the reason that up quarks, down quarks, charged leptons, and neutrinos need not have the same mass stretching.
Y_f	A trial version of the complete family mass pattern before its raw overlaps are converted into masses.
$\Pi_0 Y_f$	The part of Y_f that remains unchanged after every possible family orientation is averaged. When it exists, it acts as a steady background and a fixed ruler.
$Q_0 Y_f = Y_f - \Pi_0 Y_f$	The part that changes with family orientation. It is the lobed or displaced part measured against the steady background.
b	A trial amount of the background-versus-wave, or radial, channel. It is used to compare possible length rules; setting $b = 0$ does not remove the physical mass wave.
X	A trial turn of the three-generation pattern. It is not electric charge and not ordinary spin.
$\ell_{R_f}(X)$	The total change produced when the family turn X acts on all pieces of the sector's pattern.
$b^2 \ Q_0 Y_f\ ^2$	The radial energy cost: how much changing wave must be carried relative to a steady background.
$\ \ell_{R_f}(X)\ ^2$	The turning energy cost: how much of the full sector pattern moves during one family turn.
$b^2 \ \Pi_0 Y_f\ ^2 + \ X\ ^2 = 1$	The unit constraint. It makes the two trial channels use the same total amount, preventing either from winning by simply shrinking to zero.

Table 6 continued

Term	Plain-language meaning
S_f , Eq. (20)	The complete length-selector action. It compares the radial and turning costs while enforcing the unit constraint. The lowest valid cost determines p_f^2 .
λ	The number that enforces the unit constraint. At the selected solution it equals p_f^2 .
$\kappa_f = T(R_f)/T(\mathbf{3})$	How much family-turn response the entire sector pattern has compared with one basic family component. The model gives turning costs 6 for an endomorphism pattern, 5 for the up pattern, and 6 for the neutrino root pattern.
Eq. (23)	The ground-channel rule. If a steady background exists, compare the radial and turning costs and choose the lower one. If no steady background exists, only the turning cost can be normalized.

G.3 Why there are three different masses in each sector

The model uses one closed wave with three equally spaced family positions. At one position the changing wave nearly cancels the steady contribution, so little internal energy remains. At the second position the cancellation is weaker. At the third position the two contributions reinforce. These are the light, middle, and heavy generations.

The same particle does not rotate from electron to muon to tau, or from up to charm to top. The three generations are three separate stable versions of the same family pattern. They use the same three positions but retain different amounts of the pattern's energy. The sector values ρ_f and p_f determine how close the light state is to cancellation and how strongly the three differences are enlarged.

G.4 Why the four sectors differ

Charged leptons. The charged-lepton wave begins as two balanced, oppositely oriented components. This gives $|\rho_e| = \sqrt{2}$. Its family pattern also contains a steady background that survives every family turn. Measuring the wave against that background costs less than turning the whole pattern, so the selector chooses $p_e = 1$. The electron lies near cancellation, the muon at partial cancellation, and the tau near reinforcement. No additional stretch is needed beyond Shulga's squared overlap.

Down quarks. The quark starting balance uses the two-to-one magnitudes of the up and down charges. The cubic carrier response moves that balance to the down weight $19/54$, giving the larger wave strength $\rho_d = \sqrt{54/19}$. Like the charged-lepton pattern, the down pattern has a steady background. Its radial cost $27/19$ is below the whole-pattern turning cost 6, so $p_d = \sqrt{27/19}$. This produces a substantial but not extreme hierarchy. The light down overlap passes through zero and changes sign; the mass stays positive because the law uses its absolute size.

Up quarks. The same carrier response gives the up weight $35/54$ and hence $\rho_u = \sqrt{54/35}$. That wave strength is smaller than the down strength, but the up mass pattern joins two family pieces of the same kind. No background remains fixed under every family turn. The radial channel therefore has no invariant ruler, and the selector must turn the whole paired pattern. Some of its allowed pairings move once and some move twice, giving the combined turning cost 5 and $p_u = \sqrt{5}$. This large stretch makes the up quark extremely light relative to the top even though their raw wave strength is not unusually large.

Neutrinos. Electric charge cannot distinguish the three neutral root directions, so the model starts them with equal weights $1/3$. This gives $|\rho_\nu| = \sqrt{3}$. The links between different root directions point opposite to the self-links, which supplies the negative sign $\rho_\nu = -\sqrt{3}$ and the normally ordered pattern. The root pattern has no steady singlet background, so only the whole-pattern turning channel is available. Its six moving root directions give $p_\nu^2 = 6$. The model predicts neutrino ratios, but the unknown M_ν still sets how small all three neutrino masses are.

G.5 What this picture does not explain

The plain-language picture does not turn the proposed internal coordinates into literal objects moving in hidden space. It does not derive ordinary spin, the four overall mass scales, or the relative family orientations that produce complete CKM and PMNS mixing. Charge supplies the quark starting balance in the present model, but charge alone does not determine mass: the joining rule, three-family interference, selected length, and unknown sector scale are all separately required.

H Derivability and no-go audit of the proposed action

This appendix addresses the seven unresolved choices in the action. Here “derived” means that a result follows from equations and premises already displayed. “Conditional” means that it follows after one additional physical interpretation is stated. “Not determined” means more than that a derivation has not been found: an explicit alternative with the same stated symmetry changes the result. In that case the present assumptions cannot select the paper’s answer.

Table 7: Status of the seven action questions.

Question	Result	Status
Why is the response $A - A^3$ with unit relative coefficient?	It is the positive triple response $[B, B, A] = C^2 A = A - A^3$ of a normalized two-state carrier. The same direction is the cubic gradient of the leading E_7 quartic. Unit response at zero imbalance fixes the overall normalization.	Conditional derivation
Why are higher odd powers absent?	Lowest polynomial degree, or a tree-level leading-invariant truncation, removes them. Symmetry alone does not: $A(1 - A^2)(1 + \eta A^2)$ obeys the same elementary conditions for every η .	Not determined
Why is $ \rho = w^{-1/2}$?	It is exactly the unit-vector equation $w \rho ^2 = 1$ if w is the metric coefficient of one wave coordinate. A two-residue interpretation would instead give a different power.	Conditional derivation
Why use one unweighted trace metric?	Tensor products inherit it from one normalized family triplet. $SU(3)$ alone permits independent positive weights on inequivalent components and sectors.	Not determined
Why are radial-rotation cross terms absent?	An independent parity or a product-metric geometry forbids them. $SU(3)$ alone permits a cross term after the sector background is included.	Not determined
Why choose the lowest stationary channel?	The selector is a generalized Rayleigh quotient. Its lowest stationary value is forced if the observed state is the stable ground mode of a positive internal operator.	Conditional derivation
Why is the overlap a physical fermion mass?	An explicit two-profile internal transport action gives the mass matrix $M_f Z_f ^{2L_f}$. Its singular values are the four-dimensional fermion masses. This proves existence of a completion, not uniqueness.	Conditional completion

H.1 The unit cubic response

Consider the normalized two-state carrier used for the quark transfer. Its rank-one projector can be written

$$P = \frac{1}{2}(I + A + B), \quad A = D\sigma_z, \quad B = C\sigma_x, \quad D^2 + C^2 = 1. \quad (66)$$

The diagonal matrix A records the charge imbalance and the off-diagonal matrix B records the coherence required for P to be pure. For matrices, the Hermitian Jordan triple product is

$$[X, Y, Z] = \frac{1}{2}(XY^\dagger Z + ZY^\dagger X). \quad (67)$$

Because $B^2 = C^2 I$ and $A^3 = D^2 A$,

$$\boxed{[B, B, A] = C^2 A = (1 - D^2)A = A - A^3.} \quad (68)$$

The map $Z \mapsto [B, B, Z]$ is positive in the trace metric, since

$$\langle Z, [B, B, Z] \rangle = \frac{1}{2} \text{tr} \left(Z^\dagger B B^\dagger Z + Z^\dagger Z B^\dagger B \right) \geq 0. \quad (69)$$

This chooses the positive orientation over the equally familiar quadratic Jordan operation $BAB = -C^2 A$.

The response also has an exceptional-algebra parent. In the Freudenthal realization of the 56 of E_7 , the quartic invariant I_4 contains the exceptional Jordan cubic and quartic structures [13, 14]. On the normalized charge corner used in Eq. (66), direct differentiation gives

$$-\frac{1}{6} \nabla_A I_4(0, A, B, 0) = C^2 A = A - A^3. \quad (70)$$

Thus the relative coefficient between A and A^3 is not fitted once the standard quartic and the normalized projector are chosen. An overall multiple $\kappa(A - A^3)$ is still allowed. The boundary condition

$$D\mathcal{F}(0) = I \quad (71)$$

says that an infinitesimal imbalance is transmitted without an arbitrary rescaling and fixes $\kappa = 1$. This is a normalization premise, not a consequence of E_7 alone.

H.2 Why symmetry does not exclude higher powers

The elementary requirements used above are oddness in A , vanishing at zero coherence, unit linear response at $A = 0$, and vanishing response at the pure diagonal endpoints $A^2 = I$. For every real number η ,

$$\mathcal{F}_\eta(A) = A(1 - A^2)(1 + \eta A^2) \quad (72)$$

obeys all four requirements. It changes the quark response by the factor $1 + \eta/9$ because $D^2 = 1/9$. This is an explicit counterexample to exact uniqueness.

The same issue appears in the exceptional formulation. Replacing a leading quartic action by

$$I_4 + \frac{\alpha}{\Lambda^4} I_4^2 + \dots \quad (73)$$

preserves E_7 but rescales its gradient on the fixed charge corner. A single algebraic 56 has a distinguished quartic invariant, but a compact physical field, multiple copies, or mixed fields permit further Hermitian and mixed invariants. Exact E_7 symmetry therefore protects the response direction more strongly than its magnitude. Higher terms are absent only after a leading-order premise is added, such as choosing the lowest-degree invariant or a tree-level renormalizable truncation. Their absence cannot be derived from the present symmetry statement.

H.3 The inverse-square wave strength

Let one complex internal coordinate ξ have line element

$$ds^2 = w d\xi^* d\xi, \quad w > 0. \quad (74)$$

A unit vector in that coordinate satisfies

$$1 = w|\xi|^2, \quad |\xi| = w^{-1/2}. \quad (75)$$

Identifying ξ with the relative wave coordinate ρ_f proves Eq. (15). No fitting or further algebra is involved.

What remains conditional is the identification of a carrier weight with the metric of one coordinate. If w instead normalized each of two independent left and right endpoint residues, each endpoint factor would be $w^{-1/2}$ and their product would scale as w^{-1} . More general nonlinear field redefinitions can produce still other powers. The data do not decide which object P_f measures. The model therefore needs the single-coordinate metric interpretation in Eq. (74).

H.4 The common metric and its counterexample

Fixing one inner product on the family triplet V induces the usual tensor product, dual, symmetric-tensor, endomorphism, and adjoint metrics. Under that construction the trace ratios in Eq. (21) are fixed: the numbers 5 and 6 are representation invariants, not adjustable normalizations. This provides a reason for using the unweighted trace metric if all four sector tensors descend from one canonically normalized parent triplet.

$SU(3)$ invariance by itself is weaker. For $\text{End}(V) = \mathbf{1} \oplus \mathbf{8}$, the most general diagonal invariant norm already contains two positive constants:

$$\|Y\|_{\alpha,\beta}^2 = \alpha\|\Pi_0 Y\|^2 + \beta\|Q_0 Y\|^2, \quad \alpha, \beta > 0. \quad (76)$$

The radial candidate then changes from $\rho_f^2/2$ to

$$(p_f^{\text{rad}})^2 = \frac{\beta}{\alpha} \frac{\rho_f^2}{2}. \quad (77)$$

All values of β/α preserve $SU(3)$. Independent kinetic normalizations of the $\bar{\mathbf{6}}$ and $\mathbf{8}$ sectors similarly alter comparisons with the parent triplet. A larger symmetry or a single parent kinetic term could set these constants, but the representation list in Eq. (18) does not. The common unweighted metric is therefore a separate unification premise.

H.5 Cross terms and the role of a grading

After normalizing the radial and rotation reference coordinates, a general two-channel quadratic selector has the reduced form

$$K = \begin{pmatrix} r & c \\ c & k \end{pmatrix}, \quad \lambda_{\pm} = \frac{r + k \pm \sqrt{(r - k)^2 + 4c^2}}{2}. \quad (78)$$

Here r and k are the radial and rotation costs. The paper uses $c = 0$, so the ground value is $\min(r, k)$. For any nonzero allowed c , the lower value moves continuously away from that result.

Once a sector pattern supplies an octet direction, $SU(3)$ permits an invariant contraction between that direction and the adjoint rotation coordinate. It therefore does not force $c = 0$. Either an independent $\mathbb{Z}_2$ grading under which the two coordinates have opposite parity, or an internal product geometry in which radial and angular velocities are orthogonal, sets $c = 0$. The former is the premise used in the main text. Until that grading is obtained from a parent theory, absence of the cross term is not derived.

H.6 Why the lower stationary channel is selected

Equation (20) is a generalized Rayleigh problem. In finite-dimensional notation it has the form

$$\mathcal{S}(u, \lambda) = \frac{1}{2}u^\dagger K u - \frac{\lambda}{2}(u^\dagger N u - 1), \quad K \geq 0, \quad N > 0. \quad (79)$$

Its stationary values solve $Ku = \lambda Nu$. The Rayleigh–Ritz identity then gives

$$\min_{u^\dagger N u = 1} u^\dagger K u = \lambda_{\min}. \quad (80)$$

Consequently the lower value in Eq. (23) is forced if K is the positive internal energy operator and the observed fermion uses its stable ground channel. The higher stationary solution is then an internal excitation. Stationarity alone does not select between them; the ground-state identification is the required physical premise.

H.7 An explicit bridge from overlap to fermion mass

The remaining question is whether $|z|^{2p}$ can be more than a numerical weight. Let Z_f be the family operator in Eq. (5) and define its positive absolute value

$$|Z_f| = (Z_f^\dagger Z_f)^{1/2}. \quad (81)$$

Introduce left- and right-handed internal profiles on a dimensionless path $0 \leq s \leq L_f$ and take

$$S_{\text{prof}} = \frac{1}{2} \int_0^{L_f} ds \left[\|\partial_s \Psi_L - \log |Z_f| \Psi_L\|^2 + \|\partial_s \Psi_R - \log |Z_f| \Psi_R\|^2 \right]. \quad (82)$$

The zero-action profiles obey

$$\Psi_L(L_f) = |Z_f|^{L_f} \Psi_L(0), \quad \Psi_R(L_f) = |Z_f|^{L_f} \Psi_R(0). \quad (83)$$

After the left–right gauge bridge supplies the generation-independent scale M_f , place its bilinear at the far endpoint:

$$\mathcal{L}_{\text{bridge}} = -M_f \overline{\Psi_L(L_f)} \Psi_R(L_f) + \text{h.c.} \quad (84)$$

Substitution of Eq. (83) gives the four-dimensional mass matrix

$$\boxed{\mathcal{M}_f = M_f |Z_f|^{L_f} |Z_f|^{L_f} = M_f |Z_f|^{2L_f}}. \quad (85)$$

In the eigenbasis of Z_f , the coefficient of $\overline{\psi_{L,k}} \psi_{R,k}$ is $M_f |z_{f,k}|^{2L_f}$. The nonnegative singular values of this left–right bilinear are the tree-level fermion masses. Dimensional reduction of an internal Dirac operator is known to produce massive four-dimensional spinors [15], and extra-dimensional profile models provide explicit examples in which internal localization produces power-law fermion mass hierarchies [16]. Equation (82) is not claimed to be either cited model; it is an existence proof tailored to the present overlap operator.

This closes the logical gap between an internal overlap and a mass coefficient. It does not derive the internal path, its boundary conditions, or the scale M_f , and other transport actions could give other functions of Z_f . The present algebra therefore permits a physical mass completion but does not uniquely require it.

H.8 Minimal premises and remaining boundary

The proposed action becomes a mathematically closed effective model after six physical premises are stated:

1. use the positive leading cubic response of the normalized carrier, with unit response at zero imbalance;
2. truncate higher invariant responses;
3. interpret each carrier entry as the metric coefficient of one canonically normalized wave coordinate;
4. induce all sector metrics from one normalized family triplet and impose the parity that separates radial from rotation coordinates;
5. identify the observed channel with the ground mode of the positive selector;
6. use the internal-profile bridge of Eq. (82), or a dynamically equivalent completion, to turn the endpoint overlap into a left–right fermion mass.

Premises 1, 3, 5, and 6 lead to exact consequences once adopted. Premises 2 and 4 cannot be obtained from the symmetries currently used: Eqs. (72), (76), and (78) are counterexamples. This is the boundary of the present derivation. A deeper theory can cross it only by adding a selection principle—for example a common parent kinetic term, a derived grading, and a controlled low-energy expansion. More algebraic manipulation of the existing $SU(3)$ representations cannot do so.

This audit does not settle the separate choices of the initial carrier weights, the neutrino orientation, the four scales M_f , or the relative family frames that generate full mixing. Those remain the next physical questions after the action itself.

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